

DATA SCIENCE CAPSTONE PROJECT

HOUSE PRICE PREDICTION

Abstract

This study presents a comprehensive machine learning pipeline for predicting house prices using structured housing data. The workflow integrates preprocessing, feature engineering, feature selection, model comparison, hyperparameter tuning, and deployment. Multiple regression models were evaluated, including linear and ensemble approaches, with GridSearchCV used for systematic optimization. The final pipeline was saved for reproducibility and deployed for interactive and batch predictions. Results demonstrate strong predictive performance and highlight the practical applications of machine learning in real estate valuation, investment analysis, and urban planning.

Introduction

Accurate house price prediction is a critical challenge in real estate markets, influencing buyers, sellers, investors, and policymakers. Traditional valuation methods often rely on limited features or subjective assessments, whereas machine learning offers a data-driven approach capable of capturing complex relationships among property attributes.

This project aims to design and evaluate a robust predictive system for house prices. The objectives are:

1. To preprocess and structure raw housing data for modeling.
2. To evaluate multiple regression models with feature selection and hyperparameter tuning.
3. To deploy the best model for real-world usability.

Methodology

Data Preprocessing

The dataset was cleaned by imputing missing values (median for numerical, mode for categorical), removing duplicates, and engineering a new feature, HouseAge, derived from construction year. Numerical features were standardized, while categorical features were one-hot encoded.

Feature Selection

Feature relevance was assessed using SelectKBest with statistical tests, reducing dimensionality and improving interpretability.

Model Development

Five models were compared: Linear Regression, Ridge Regression, Lasso Regression, Random Forest Regressor, and Gradient Boosting Regressor. Hyperparameters were tuned using GridSearchCV with cross-validation, optimizing parameters such as regularization strength, number of trees, and maximum depth.

Deployment

The best pipeline was saved using Pickle, enabling reproducibility and future predictions. Interactive input and batch prediction modes were implemented to support diverse use cases.

Data Preprocessing report

1. Data Loading

- The dataset was imported from a CSV file containing house attributes (Area, Bedrooms, Bathrooms, Floors, YearBuilt, Location, Condition, Garage) and the target variable (Price).
- Initial inspection (head(), info(), describe()) confirmed the presence of both numerical and categorical features, with no major structural issues.

2. Data Cleaning

- Missing values:
 - Numerical columns → replaced with median values to reduce sensitivity to outliers.
 - Categorical columns → replaced with mode values to preserve the most frequent category.
- Duplicates: Removed to avoid bias and redundancy in training.

3. Feature Engineering

- A new feature HouseAge was created.
- This captures the age of the property, which is often a strong predictor of price.

4. Feature Separation

- Predictors (X): All columns except Price (target) and Id (identifier).
- Target (y): Price.

5. Feature Categorization

- Categorical features: Location, Condition, Garage.
- Numerical features: Area, Bedrooms, Bathrooms, Floors, YearBuilt, HouseAge.

6. Preprocessing Strategy

- Numerical features: Standardized using StandardScaler to ensure all variables are on a comparable scale.
- Categorical features: Encoded using OneHotEncoder with handle_unknown='ignore' to convert categories into binary indicators.

7. Pipeline Construction

- A ColumnTransformer was defined to apply numerical scaling and categorical encoding simultaneously.
- This was wrapped into a Pipeline for reproducibility and modularity.

8. Transformation Results

- Before preprocessing: Shape = (n_samples, n_features) with mixed data types.
- After preprocessing: Shape expanded due to one-hot encoding of categorical variables.
- The transformed dataset is now fully numeric and suitable for machine learning models.

Modeling & Evaluation Report

1. Feature Selection

- Applied SelectKBest with f_regression to identify the most statistically significant predictors of house price.
- This step reduces dimensionality, improves interpretability, and helps avoid overfitting.
- Multiple values of k (number of features selected) were tested (10, 20, 50, and all) to evaluate the impact of feature subset size on model performance.

2. Model Candidates

Several regression models were considered to capture both linear and non-linear relationships:

- Linear Regression – baseline model for interpretability.
- Ridge Regression – adds L2 regularization to handle multicollinearity.
- Lasso Regression – adds L1 regularization for feature selection.

- Random Forest Regressor – ensemble of decision trees, robust to non-linearities.
- Gradient Boosting Regressor – sequential ensemble learning, effective for complex patterns.

3. Hyperparameter Tuning

- Used GridSearchCV with 5-fold cross-validation to systematically explore hyperparameters.
- Examples of tuned parameters:
 - Ridge/Lasso: alpha (regularization strength)
 - Random Forest: n_estimators, max_depth
 - Gradient Boosting: n_estimators, learning_rate, max_depth
- Evaluation metric: R^2 score (coefficient of determination), chosen for its interpretability in regression tasks.

4. Best Model Selection

- GridSearchCV identified the best combination of model + hyperparameters + feature subset size.
- The pipeline ensured preprocessing, feature selection, and modeling were applied consistently across all folds.
- The best model achieved the highest cross-validation R^2 score and was validated on the held-out test set.

5. Model Persistence

- The final best model (including preprocessing and feature selection steps) was saved using Pickle for future reuse.
- This ensures reproducibility and allows predictions on new unseen data without retraining.

6. Evaluation Results

- Cross-validation score (R^2): Indicates how well the model generalizes across folds.
- Test score (R^2): Confirms performance on unseen data.
- The chosen model demonstrated strong predictive ability, balancing accuracy and interpretability.

Results

- Preprocessing produced a fully numeric dataset suitable for regression modeling.
- Feature selection identified subsets of predictors that improved model efficiency.
- Model evaluation showed ensemble methods (Random Forest, Gradient Boosting) outperforming linear models in terms of R^2 score.
- Best model achieved strong generalization on unseen test data, confirming robustness.
- Deployment allowed seamless predictions on new inputs, with outputs exportable for reporting.

Discussion

The results highlight the importance of combining preprocessing, feature engineering, and model optimization in predictive analytics. Ensemble methods captured non-linear relationships more effectively than linear approaches, though regularized linear models offered interpretability advantages.

The deployment strategy ensures practical usability, allowing both individual users and organizations to benefit from automated predictions. The system demonstrates how machine learning can enhance decision-making in real estate, investment, and policy contexts.

Future work could incorporate additional contextual features (e.g., neighborhood amenities, proximity to transport), explore advanced algorithms (XGBoost, LightGBM, deep learning), and integrate explainability frameworks (SHAP, LIME) to improve transparency.

Conclusion

1. End-to-End Workflow

This project successfully demonstrated a complete machine learning pipeline for house price prediction:

- Preprocessing: Cleaning, handling missing values, feature engineering (HouseAge), scaling, and encoding.
- Modeling & Evaluation: Feature selection with SelectKBest, model comparison (Linear, Ridge, Lasso, Random Forest, Gradient Boosting), and hyperparameter tuning via GridSearchCV.
- Deployment & Usage: Saving the best pipeline with Pickle, interactive user input for predictions, and batch prediction/export for reporting.

2. Key Achievements

- Built a robust, reproducible pipeline using scikit-learn.
- Identified the best-performing model with strong generalization on unseen data.
- Enabled real-world usability by saving and reloading the model for predictions.
- Provided both interactive and batch prediction modes, making the system versatile.

3. Practical Applications

- Real Estate Valuation: Assists agents and buyers in estimating fair property prices.
- Investment Analysis: Helps investors identify undervalued or overvalued properties.
- Urban Planning & Policy: Provides insights into how features like location, condition, and age affect housing markets.
- Academic & Research Use: Serves as a reproducible case study in applied machine learning for regression tasks.

4. Future Enhancements

- Incorporate additional features (e.g., proximity to schools, transport, amenities).
- Explore advanced models (XGBoost, LightGBM, Neural Networks).
- Add explainability tools (SHAP, LIME) to interpret predictions.
- Deploy as a web app or API for broader accessibility.

This project demonstrates how structured preprocessing, careful model selection, and thoughtful deployment can transform raw housing data into a powerful predictive tool. It bridges data science rigor with real-world usability, making it valuable for both academic research and industry applications.